

Complete Viva Questions & Answers (Hinglish)

Topic: Video Tracking, Kalman Filter, KF-based Trackers & Deep Learning Filters

PAPER 1: Kalman Filter–Based Tracking for Fast-Moving Tiny Objects

Q: Is paper ka objective kya hai?

A: Is paper ka objective yeh evaluate karna hai ki existing Kalman Filter based trackers fast-moving aur tiny objects (jaise racquetball) ko accurately track kar paate hain ya nahi.

Q: Tiny object tracking difficult kyun hota hai?

A: Object chhota hota hai isliye pixels kam milte hain, motion blur aata hai, aur object har frame mein large displacement karta hai. Yeh sab KF ke linear motion assumption ko tod dete hain.

Q: Is paper mein kaunsa tracking paradigm use hua hai?

A: Tracking-by-detection paradigm use hua hai jisme har frame object detect hota hai aur tracker un detections ko frame-to-frame associate karta hai.

Q: Is paper mein kaunse trackers evaluate kiye gaye?

A: SORT, ByteTrack, StrongSORT, OC-SORT jaise Kalman Filter–based trackers evaluate kiye gaye.

Q: Kalman Filter yahan fail kyun karta hai?

A: KF assume karta hai motion smooth aur linear hoga, jabki racquetball ka motion highly non-linear aur unpredictable hota hai.

Q: Paper ka main conclusion kya hai?

A: Sabhi trackers mein spatial error bahut zyada aata hai (3–11 cm), jo practical applications ke liye unacceptable hai.

PAPER 2: Beyond Kalman Filters – Deep Learning Based Filters

Q: Is paper ne Kalman Filter ko replace karne ki baat kyun ki?

A: Kyuki KF manually tuned hota hai aur non-linear motion ko handle nahi kar paata. Deep learning models motion ko data se seekh sakte hain.

Q: Filter-level learning ka matlab kya hai?

A: Filter-level learning ka matlab sirf motion prediction aur correction part ko neural network se replace karna, poore tracker ko nahi.

Q: End-to-end filter kya hota hai?

A: End-to-end filter wo hota hai jisme prediction aur update dono neural network seekhta hai, aur measurement noise manually define nahi karna padta.

Q: Neural ODE ka use kyun kiya gaya?

A: Neural ODE continuous-time motion ko model karta hai, missing detections handle karta hai aur non-linear trajectories ko achhe se represent karta hai.

Q: RNN-CNP ka main fayda kya hai?

A: RNN-CNP sparse aur noisy data mein bhi smooth trajectory predict kar sakta hai, isliye occlusion mein better perform karta hai.

Q: MoveSORT kya hai?

A: MoveSORT ek improved SORT tracker hai jisme Kalman Filter ko deep learning filter se replace kiya gaya hai aur matching cost bhi improve ki gayi hai.

Q: IoU + L1 distance kyun use kiya gaya?

A: Sirf IoU object ke exact position aur scale difference ko capture nahi karta, isliye L1 distance add karke matching better banayi gayi.

Q: Deep learning slow hone ke bawajood kyun useful hai?

A: Kyuki accuracy, robustness aur occlusion recovery significantly improve hoti hai, aur real-time constraints abhi bhi satisfy ho jaate hain.

PAPER 3: Kalman Filter–Based Trackers (SORT Family)

Q: Tracker-level aur filter-level mein kya difference hai?

A: Tracker-level poora system hota hai jo detection, matching aur ID management karta hai, jabki filter-level sirf state estimation karta hai.

Q: SORT tracker ka basic workflow kya hai?

A: Detection → KF prediction → IoU based cost matrix → Hungarian algorithm → track update.

Q: Hungarian algorithm ka role kya hai?

A: Hungarian algorithm optimal matching find karta hai detections aur predicted tracks ke beech.

Q: OC-SORT ya High-SORT kya improve karte hain?

A: Ye trackers better occlusion handling, direction consistency aur track recovery add karte hain, lekin abhi bhi Kalman Filter use karte hain.

Q: Kalman Filter ka main advantage kya hai?

A: Bahut fast hai, computationally efficient hai, aur real-time applications ke liye suitable hai.

Q: Kalman Filter ki limitation kya hai?

A: Non-linear motion, sudden acceleration aur noisy detections ke case mein performance degrade ho jaati hai.

Q: Is paper se project ke liye kya seekhne ko milta hai?

A: Kalman Filter ek strong baseline hai, lekin modern applications ke liye learning-based improvements future direction hain.